

FINAL EXPLANATION: PHYSICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: PHYSICS GRADE 10 - PRESSURE

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

You have finished the unit on pressure. Now you will write a Final Explanation to answer our Driving Question: "Why did the jerrycan crush itself, and how does the invisible push of fluids move mountains, save lives, and power machines across Kenya?" Answer each section in the space under its prompt. Use evidence from your lessons, your Driving Question Board, your cumulative model and your calculations. Where a prompt asks for a calculation, show the formula, the substitution and the unit. Where a prompt asks for a diagram, label every arrow. Write in full sentences and use the words: pressure, force, area, atmospheric pressure, density, depth, Pascal's Principle.

Section 1: Why Did the Jerrycan Crush Itself?

In Lesson 1 you watched a sealed, empty 20-litre jerrycan crumple inward after a small amount of air was removed from it, while nobody touched it. Write a complete explanation of why it collapsed. (a) Describe what the air inside and outside the jerrycan was doing BEFORE air was removed and say why the jerrycan stayed intact. (b) Explain what changed when air was removed. (c) A jerrycan is strong enough to carry 20 kg of water every day without deforming, yet this sealed, empty jerrycan collapsed. Explain how both facts can be true. (d) Draw and label a pressure-balance diagram showing the jerrycan before and after air removal. (e) Use evidence from the collapsing tin (Lesson 4), the Magdeburg Hemispheres and the mercury barometer.

Write your answer here:

Section 2: Why Don't Our Bodies Get Crushed?

Atmospheric pressure acts on your body all the time, yet you are not crushed. (a) Using what you learned about pressure balance, explain why not. (b) Explain what is the same and what is different between your body and the jerrycan after air was removed. (c) Explain why atmospheric pressure is described as acting in ALL directions, and give one piece of evidence from our lessons.

Write your answer here:

Section 3: Pressure in Liquids and Kenyan Structures

In Lessons 3, 6 and 8 you investigated liquid pressure. (a) State the three factors that control liquid pressure and describe the evidence from the water-jet bottle investigations that showed how depth and density matter. (b) Write the formula and calculate the pressure at the base of the Sasumua Dam reservoir (depth about 40 m, fresh water density 1000 kg/m^3 , $g = 10 \text{ N/kg}$). (c) Calculate the pressure due to the seawater on a diver at 10 m depth in the Indian Ocean off Mombasa (seawater density 1025 kg/m^3). (d) Compare your answers in (b) and (c) with atmospheric pressure. What does the comparison tell you about the design of the Sasumua Dam wall?

Write your answer here:

Section 4: Pascal's Principle and Hydraulic Machines

In Lessons 5 and 7 you used connected syringes and watched a hydraulic car lift in a Nairobi garage. (a) State Pascal's Principle in your own words. (b) Explain how a small force on a small piston can lift a heavy vehicle such as a Toyota Noah. (c) A small piston has an area of 5 cm^2 and you push it with a force of 40 N. The large piston has an area of 500 cm^2 . Calculate the force on the large piston. (d) Name two other Kenyan hydraulic machines we studied. (e) Explain why removing air bubbles from the syringe system was important.

Write your answer here:

Section 5: Putting It All Together

Write a final answer to the Driving Question: "Why did the jerrycan crush itself, and how does the invisible push of fluids move mountains, save lives, and power machines across Kenya?" (a) Say how the jerrycan collapse is DIFFERENT from a hydraulic machine, and say what they have in common. (b) Choose TWO Kenyan examples (for instance a dam, a diver, a matatu brake, a syringe or the Kenya Meteorological Department barometer) and explain which pressure idea each uses. (c) Name one question from our Driving Question Board that you can now answer, and one that is still open.

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Explaining the anchoring phenomenon (jerrycan)	Clearly explains the pressure imbalance before and after air removal, uses P_{outside} and P_{inside} , links to net force ($F = \Delta P \times A$), and uses a correct, fully labelled diagram with evidence from the tin, the Magdeburg Hemispheres and the barometer.	Explains that outside pressure became greater than inside pressure and the jerrycan was crushed, with a diagram and some evidence, but with minor gaps in reasoning or labelling.	States that the air crushed the jerrycan without explaining the pressure difference or the balanced state before, or gives little or no evidence.
Use of formulas and calculations	Correctly selects $P = F/A$, $P = \rho gh$ and $F_1/A_1 = F_2/A_2$, shows substitution and units, and obtains 400,000 Pa, 102,500 Pa and 4,000 N.	Uses the correct formulas with mostly correct answers, with minor errors in units or in showing the working.	Chooses incorrect formulas or makes major errors, or gives answers without working or units.
Use of evidence from lessons	Draws specifically on the water-jet investigations, the collapsing tin, the Magdeburg Hemispheres, the barometer, the syringes and the car lift to support each claim.	Refers to some lesson evidence to support most claims.	Makes claims with little or no reference to lesson evidence.
Connecting concepts and Kenyan applications	Distinguishes uncontrolled pressure difference from controlled pressure transmission, links atmospheric pressure,	Links most concepts and explains at least one Kenyan example accurately.	Treats ideas separately, with applications that are missing or inaccurate.

	liquid pressure and Pascal's Principle, and explains two Kenyan examples accurately.		
Scientific communication	Writes clear, logical, complete sentences with correct terms (pressure, force, area, density, depth, Pascal's Principle) and reflects thoughtfully on the Driving Question Board.	Writes generally clear explanations with mostly correct scientific terms.	Writing is unclear or incomplete, with frequent misuse of terms.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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